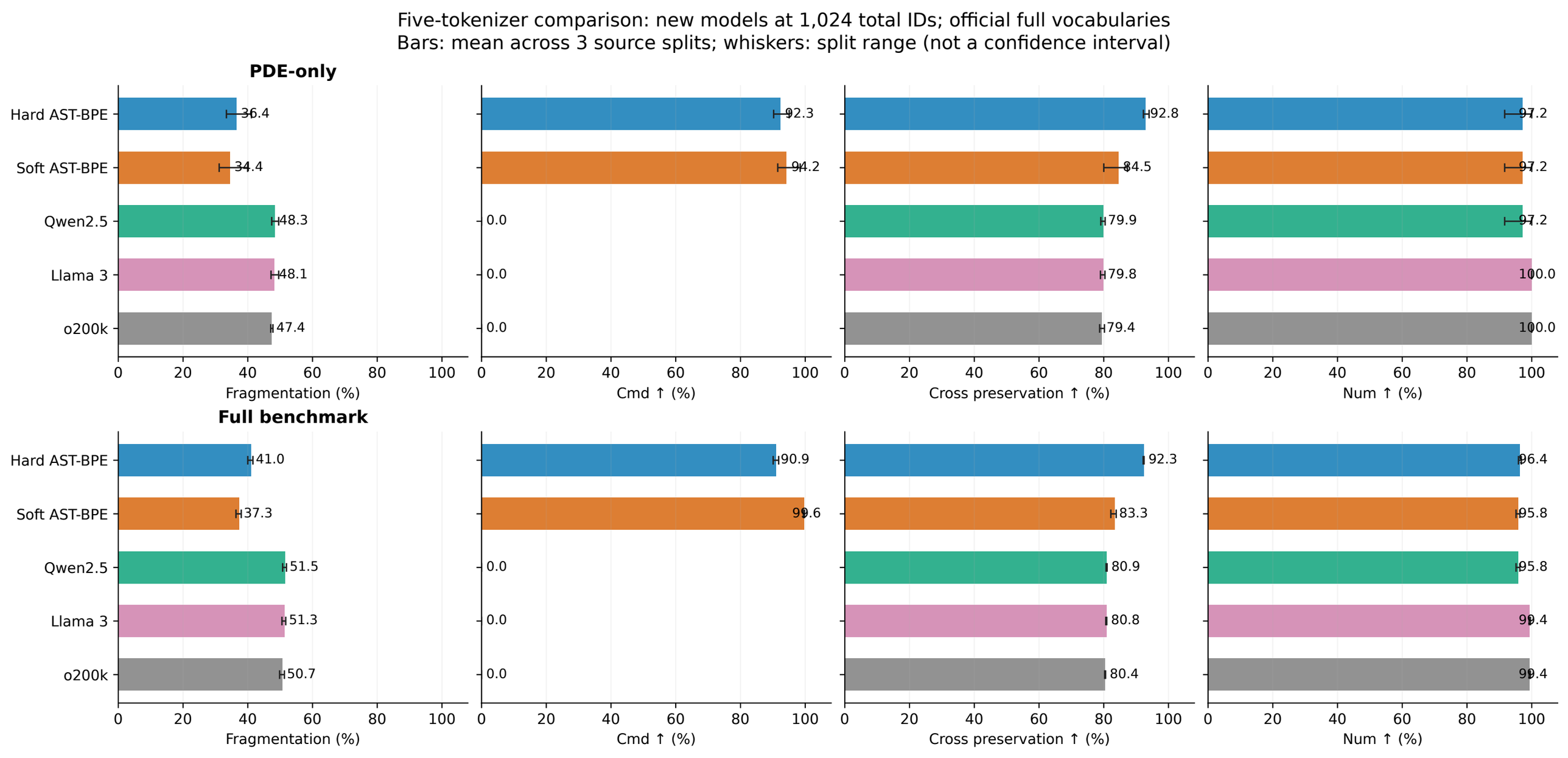
Five-tokenizer comparison: new models at 1,024 total IDs; official full vocabularies
Bars: mean across 3 source splits; whiskers: split range (not a confidence interval)
PDE-only
Full benchmark
Hard AST-BPE
Soft AST-BPE
Qwen2.5
Llama 3
o200k
36.4
34.4
48.3
48.1
47.4
92.3
94.2
0.0
0.0
0.0
92.8
84.5
79.9
79.8
79.4
97.2
97.2
97.2
100.0
100.0
41.0
37.3
51.5
51.3
50.7
90.9
99.6
0.0
0.0
0.0
92.3
83.3
80.9
80.8
80.4
96.4
95.8
95.8
99.4
99.4
0
20
40
60
80
100
Fragmentation (%)
Cmd ↑ (%)
Cross preservation ↑ (%)
Num ↑ (%)